

Dear Editor,

We are pleased to submit our manuscript entitled “Lorenz Energy Cycle Climatology for the Southwestern Atlantic Cyclones” for consideration for publication in *Climate Dynamics*.

This study presents the first comprehensive climatology of the Lorenz Energy Cycle (LEC) applied to cyclonic systems in the Southwestern Atlantic. We employ a semi-Lagrangian approach — rarely used in climatological LEC studies — to better isolate the energy dynamics of individual cyclones by following their trajectories, reducing the interference from surrounding synoptic features.

A major innovation of this work lies in the objective separation of cyclone life cycle phases using an objective and automated classification scheme. This novel methodology enables the diagnosis of energy conversions and budget terms throughout different development stages, allowing a more precise understanding of the dynamical mechanisms driving these systems.

Our results highlight the crucial role of barotropic conversions — often overlooked in the literature — in the evolution and maintenance of cyclonic activity in this region. Additionally, we explore dominant modes of variability in cyclone energetics through Empirical Orthogonal Function (EOF) analysis, and apply unsupervised clustering techniques to the most intense systems, opening new perspectives for classifying and grouping cyclones based on their energetics.

We believe these findings represent a significant advancement in understanding the dynamics of extratropical cyclones and their energy cycles in the South Atlantic. The insights offered by this study may also support the development of new classification schemes and predictive tools for cyclones, with potential applications in operational meteorology and climate modeling.

We appreciate your consideration of our manuscript for publication in *Climate Dynamics* and look forward to your feedback.

Sincerely,

Danilo Couto de Souza,

on behalf of all co-authors

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